Superpermeability of solid membranes and gas evacuation

Part I. Theory

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Solid membrane permeable for a substantial part of the incident gas flow in one direction and impermeable in the inverse direction might be used as an ideal vacuum pump. In the case where gas is absorbed without dissociation and the gas temperature is equal to that of the membrane, only an infinitesimal part of the incident gas flow can penetrate through the membrane. If, however, the gas is 'hotter' than the membrane (on account of excitation, dissociation, heating, etc.) the membrane permeability may become comparable to that of the opening in a thin wall (superpermeability). The condition indispensable for this to take place is the inlet boundary special state hindering the thermal desorption of the absorbed gas. After special boundary processing the membrane becomes superpermeable for non-equilibrium gas while ordinary permeability, small as it is, becomes practically negligible. So when non-equilibrium is maintained only at one side of the membrane, gas would flow only in one direction as is required for pumping, compression and recuperation. In the case where the absorption is accompanied by dissociation (as in the H_2–Me systems) superpermeability in principle may be achieved with equal temperatures of gas and the membrane. However, in the temperature and pressure ranges of practical interest this can be achieved only in the H_2–Pd system and with a clean membrane surface. Under the non-equilibrium condition superpermeability is possible for various gas–solid combinations over a wide range of the gas pressures, membrane temperatures (low temperatures included) and boundary conditions.

1. Introduction

It is obvious that the very existence of vacuum technology is only possible due to the low permeability of solid membranes used as vacuum device walls. From this point of view the very idea to use the gas permeability of solids for the construction of an effective means of evacuation seems rather unusual. Imagine, nevertheless, that we can fabricate solid membranes with any required properties, then the assertion would hardly provoke an objection that the best conceivable pump would be a membrane which (1) is permeable for a considerable part of an incident gas flow from the volume being evacuated; (2) is not permeable for a gas flow incident on it from the other 'exhaust' side, i.e. where the evacuated gas is compressed.

Such a membrane would be a part of the wall of the evacuated device and since the membrane state does not depend on the quantity of permeated gas, the life time of the pump would be unlimited in principle, including cases of continuous gas supply. The gas compression stipulated in item (2) can be used directly for gas recovery in those vacuum systems where the evacuated gas is a working body.

It is shown in theoretical papers[1,2] that gas–solid membrane systems possessing, in particular, the aforenamed properties, are possible in principle. In further papers[3–5] this has been proved to be the case experimentally. The objective of the present exercise is to expound successively ideas and experimental results relevant to the problem of evacuation by the aforenamed method.

The paper of G Carter *et al*[16] should also be noticed, in which it is proposed to evacuate by deep implantation of particles into the bulk of the membrane and exposure of the membrane to continuous hard radiation to provide the bulk diffusion coefficient gradient. The authors, however, consider the method proposed in other papers[1–5] more reasonable, because to realize it instead of energies in the gas of $\geqslant 100$ keV[16] energies of ≈ 1 eV are sufficient (in some cases, ambient thermal energies are sufficient), and instead of continuous hard irradiation of a membrane a single treatment of a membrane surface is sufficient.

2. Analogy between porous and solid membranes

Leaving aside for the present the question of gas compression by a membrane, let us consider the question whether membrane of a usual thickness can at all let pass through itself a sufficient part of an incident gas flow to assure reasonable rates of evacuation for a unit surface.

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Let us consider at first the simplest porous membrane of thickness l, consisting of a set of straight, parallel, thin-walled channels with square sections of side d. It is well known[6] that in the molecular regime, in the case $d \ll l$, approximately $d/l \ll 1$ part of an incident flux would pass through such a membrane. In other words, the probability for gas particles to permeate such a membrane at one collision with its entrance surface is d/l. The reason for χ being $\ll 1$ is the following. The majority of gas particles enters a membrane channel near (at the distance of $\approx d$) its entrance and then moves in rectilinear jumps of mean length $d \ll l$, changing direction arbitrarily at each collision with a wall. The random character of the motion and the proximity of the entrance opening to the initial point of such motion are the reasons for the fact that only a fraction $\chi \approx d/l$ part of particles which enters the membrane permeates the latter.

But in the solid membrane absorbed particles move also in rectilinear jumps of length λ from one equilibrium position to another, each time changing arbitrarily the direction of the following jump. And as in the porous membrane, the initial point of their motion is at the entrance boundary of the membrane or in its immediate proximity. It follows from this that the gas permeability of a solid membrane can be compared with that of the hypothetical porous membrane with $d = \lambda$. For such a porous membrane $\chi \approx \lambda/l$. In most cases λ is the crystal lattice constant. For example, if $\lambda = 4 \times 10^{-8}$ cm and $l = 10^{-1}$ cm we get $\chi = 4 \times 10^{-7}$. An evacuation rate of such a membrane per unit surface area would be $\frac{1}{4} v\chi$, where v is the particle mean velocity in the gas. For example, for hydrogen at 20°C this value would be 1.7×10^{-2} cm^3/cm^2 s which is obviously quite unacceptable for the construction of a pump on the basis of such a membrane.

It seems that a solid 'entire' membrane may be less permeable than a hypothetical porous one. In fact, in the case of a solid membrane the situation is aggravated by the fact that a permeating particle needs to overcome entrance and exit boundaries and moreover, overcome a diffusion energy barrier at each jump. Here, however, it seems useful to extend the analogy between porous and solid membranes. While moving in a porous membrane gas particles may be adsorbed by channel walls and, in particular, an average velocity of the particle motion in the channel may be determined not by their thermal velocity in a free state, but the energy of their bond with the wall surface and by the wall temperature. The motion in this case would be of an activated diffusion character and from the point of view of kinetics it would be completely similar to that of absorbed particles in a solid. It is well known, however, that the conductance of a channel, pipe, porous membrane per unit area is always independent of adsorption properties of the wall material and wall temperature. It is equal to $\frac{1}{4} vd/l$ and consequently this value does not depend on a diffusion coefficient*. It follows from this analogy that the gas permeability of a solid membrane should also be independent of the bulk properties of the membrane material so only phase boundaries distinguish porous and solid membranes in terms of kinetics. However, the generally accepted expressions for solid membrane permeability include a diffusion coefficient and a heat of gas solution in a solid. In order to establish the possibility of achieving high gas permeability of solid membranes it is necessary to consider this contradiction and to show that the existence of interphase boundaries can not only forbid gas permeation but can also raise it considerably.

* It should be noted, that the time of setting in the stationary flow of permeability depends essentially on the diffusion coefficient (a classic method of determination of an adsorption heat is based on this fact).

3. Interphase processes and conditions of superpermeability of thick membranes

Here we shall continue the consideration of the question of the maximum values of solid membrane gas permeability, leaving aside, as before, the question of gas compression. Therefore, we consider the simplest case (Figure 1) when a

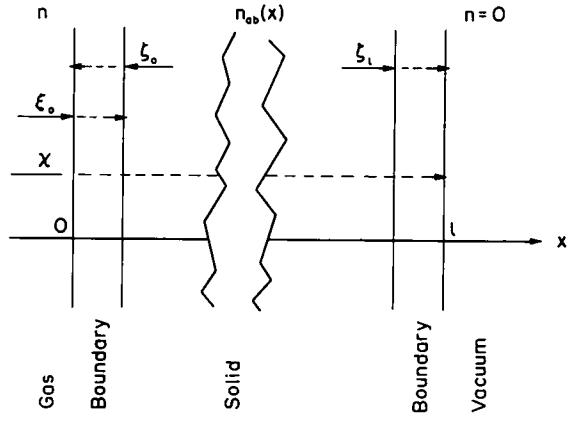


Figure 1. Membrane separating gas from a vacuum.

membrane separates gas of density n from a vacuum (in this case the volume which is being evacuated through a membrane is that containing the gas).

3.1. Deduction of an expression for permeation probability. The probability χ of gas particle permeation at a single collision with a membrane will be determined as

$$\chi = \frac{j}{i_0}, \tag{1}$$

where i_0 is the density of a flux incident on a membrane and j is the density of a stationary permeating flux. According to Fick's law

$$j = -D \frac{dn_{ab}}{dx}, \tag{2}$$

where D is the gas diffusion coefficient in the membrane material and $n_{ab}(X)$ is the stationary distribution of gas density in the membrane which will be determined from the equation

$$\frac{d^2 n_{ab}}{dx^2} = 0. \tag{3}$$

We shall consider an absorbed phase as a gas with a free-path λ and an average translation velocity v_{ab}. According to kinetic theory

$$v_{ab} = \frac{3D}{\lambda} \tag{4}$$

provided D is the same in all directions, as will be assumed in our consideration*.

We shall call the absorption probability ξ the probability that a gas particle surmounts the boundary layer at a single collision with the membrane surface.

The extension of the boundary layer is such that outside it one may neglect the influence of the surface on the gas particle state in the membrane. For a solid it is one or several lattice constants (1λ–3λ), for a channel or a porous membrane the boundary is simply a plane of an inlet surface. By analogy one can introduce the probability ζ of the absorbed particle to outlet from the membrane at a single reaching of the boundary layer (see Figure 1). In these terms one can describe the boundary conditions as follows:

$$x = 0 \quad -D\frac{dn_{ab}}{dx} = i_0\xi_0 - \tfrac{1}{4}n_{ab}v_{ab}\zeta_0,$$

$$x = l \quad -D\frac{dn_{ab}}{dx} = \tfrac{1}{4}n_{ab}v_{ab}\zeta_l. \tag{5}$$

In equation (5) '0' and 'l' indices refer respectively, to the right and left sides of the membrane (Figure 1). From equations (3) and (5) and taking account of equation (4) one gets

$$n_{ab} = \frac{4\xi_0[1 + \frac{3}{4}\zeta_l(l - x/\lambda)]}{v_{ab}[\zeta_0 + \zeta_l + \frac{3}{4}\zeta_0\zeta_l(l/\lambda)]}\, i_0 \tag{6}$$

and then taking account of equations (1) and (2)

$$j = \chi i_0 = \frac{\xi_0}{1 + (\zeta_0/\zeta_l) + \frac{3}{4}\zeta_0(l/\lambda)}\, i_0. \tag{7}$$

3.2. Determining role of boundary processes. It is clearly seen from equation (7) that the whole dependence of χ (or, which is almost the same, of j) on the membrane material, gas species and temperature is included only in the probabilities ξ_0, ζ_0, ζ_l of boundary processes. For example, by stimulating absorption and desorption one can always make ξ = constant, ζ_0 = constant, ζ_l = constant, and, consequently, obtain practically solid membrane gas permeability dependent neither on its material, nor on gas species or on temperature.

In particular, in the case of a channel or a porous membrane the processes at the boundary have always a maximum constant probability $\xi_0 = \zeta_0 = \zeta_l = 1$ and from formula (7) we get for this case the result $\chi \approx \lambda/l$ which we have already used before.

Now we shall get from equation (7) a known expression for gas permeability taking account of the processes at the phase boundary. For this assume $i_0 = \frac{1}{4}nv$ and multiply denominator and numerator of equation (7) by v_{ab}

$$j = n\frac{v\xi_0}{v_{ab}\zeta_0}\,\frac{1}{(4/v_{ab}\zeta_0) + (4/v_{ab}\zeta_l) + (3l/v_{ab}\lambda)}. \tag{8}$$

In the case of spontaneous (non-stimulated) development of boundary processes and when gas dissociation does not occur during absorption $v\xi_0/v_{ab}\zeta_0 = \Gamma$, $\frac{1}{4}v_{ab}\zeta_0 = K_0$ and $\frac{1}{4}v_{ab}\zeta_l = K_l$ where Γ is the Henry solubility constant, K_0 and K_l are rate constants of gas outlet of the membrane. Taking account of equation (4) we get from equation (8)

$$j = \frac{n\Gamma}{(1/K_0) + (1/K_l) + (l/D)}. \tag{9}$$

Reference 8 contains formulas similar to equation (9).

The expression (9) in that particular case for which it has been deduced from equation (7) is evidently equivalent to equation (7). But the very procedure of its deduction includes a superfluous operation, namely multiplication of the denominator and numerator of equation (7) by v_{ab}. After introduction of appropriate designations χ (or j) seems to depend explicitly on D (or v_{ab}) and Γ, i.e. on membrane bulk properties. Based on equation (9) one may come to an incorrect conclusion that slow diffusion in a solid always prevents gas permeation and correspondingly the problem posed in the present paper can not be solved.

3.3. Conditions of superpermeability and trapping. It is customary to consider that slow (low probability) processes at the interphase boundary, which determine in accordance with the above the whole difference in the permeability of solid and porous membranes, only reduce the gas permeability. (By the way, this follows directly from equation (9)). It is obvious, however, that formula (7) implies indication of kinetics resulting in structures which are much more permeable than a porous membrane, this being true exactly when some boundary processes have a low enough probability. Really, in the case

$$\xi_0 = 1, \tag{10}$$

$$\zeta_0 \ll \frac{\lambda}{l}, \tag{11}$$

$$\zeta_l \gtrsim \zeta_0 \tag{12}$$

we get $\chi \gtrsim \frac{1}{2}$ from equation (7), i.e. thick membrane permeability which not only exceeds that of porous membranes, but is close to that of an opening in a thin wall and we call this a superpermeability. Thus, the question is whether equations (10)–(12) are possible in real gas–solid systems? However, at first, we shall explain the kinetic mechanism resulting in a superpermeability in the case equations (10)–(12).

The conditions in equations (10)–(12) are ones of formation of a trap for gas particles: the entry into the membrane is free (equation (10)) and the outlet back into the gas is strongly hampered (equations (11)–(12)). On the average an absorbed particle should return l/λ times to the inlet boundary before it reaches the outlet boundary. However, according to condition (11) for a particle to return back into the gas requires a much larger number of attempts—the inlet boundary 'reflects strongly' absorbed particles. Consequently, taking account of equation (10) practically all the incident particles reach the outlet boundary. Let ζ_l be $\approx\zeta_0 \equiv \zeta$ for definiteness in equation (12). Then strong internal reflection occurs also at the outlet boundary and the majority of particles reaching it will return again to the inlet boundary. This occurs on the average ($1/\zeta l/\lambda \gg 1$) times. The particle 'forgets' from which side it has entered and in the case $\zeta_l \approx \zeta_0$ this results in symmetric desorption through both boundaries. Taking account of equation (10) one gets $\chi \approx \frac{1}{2}$. If in equation (12) (for example, the boundaries are not identical) the majority of absorbed particles desorbs through the inlet boundary, $\chi > \frac{1}{2}$.

* Taking account of diffusion substance anisotropy one would get instead of 3 in equation (4) a somewhat different factor[7] which will not affect the results in practice.

4. Maximum gas permeability of solid membranes in the case of absorption without dissociation in equilibrium systems*

All the monoatomic gas–solid systems and many diatomic gas–dielectric systems pertain to this case.

Let us consider two particular examples of a potential distribution in the gas–solid system: a distribution of type 1 (Figure 2) and the distribution of type 2 (Figure 3). In Figures 2 and 3 the level of the particle potential energy in the gas is accepted to be 0.

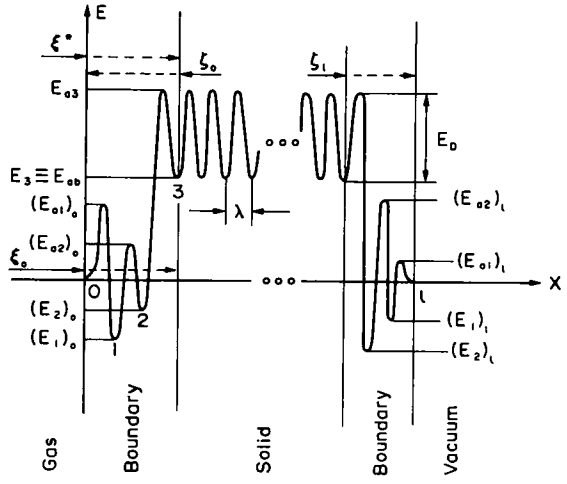


Figure 2. Potential distribution of a gas-membrane system of type 1: $E_{a1} < E_{a3}$, $E_{a2} < E_{a3}$, $E_{a3} > 0$. The type 1 distribution corresponds to the notion of an 'active' membrane.

Potential wells 1, 2, 3 correspond to the position of the particle in the state of adsorption, in the near-surface layer, and in the bulk E_1, E_2, E_3 are the levels of corresponding potential wells, whereas E_{a1}, E_{a2}, E_{a3} are the levels of corresponding potential barriers. If the membrane surface is pure (i.e. is completely accessible for adsorption–desorption) then it is easy to conclude that in systems with a distribution of type 1 (Figure 2) formation of the trap and superpermeability are impossible.

In fact, the absorbed particle impinging at the boundary (see Figure 2 and the formula (5)) is not lower than the E_{a3} level and consequently, it passes without any obstacle to position 2. From position 2 the transition is much more probable to the gas than through a high activation barrier back into solid. Thus, almost quite independently of temperature

$$\zeta_0 = \zeta_l = b \approx 0.1 - 1, \tag{12'}$$

where b is a constant of the order of magnitude 0.1–1†. Consequently, in this case it is impossible to realize the conditions of

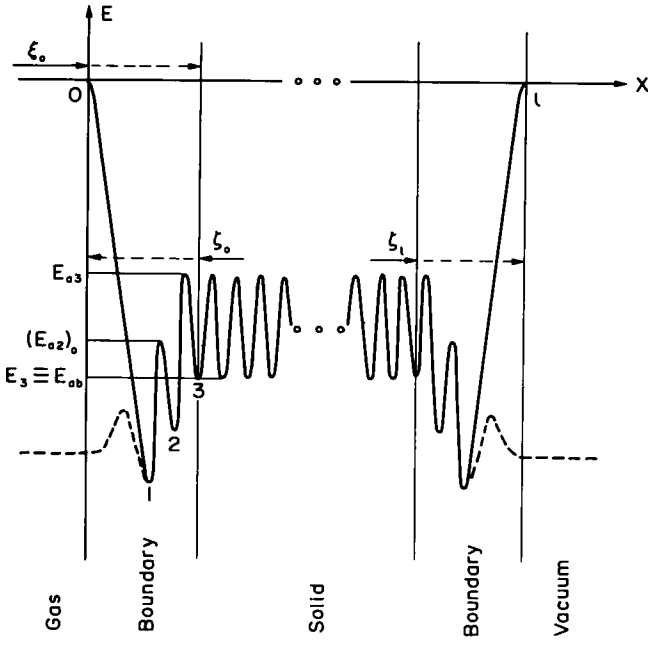


Figure 3. Potential distribution of a gas-membrane system of type 2: $E_{a1} = 0$, $E_{a2} < 0$, $E_{a3} < 0$. The type 2 distribution corresponds to an equilibrium H–Me system. The dotted line represents a distribution of the H_2–Me system.

strong internal reflection $\zeta_0 \ll \lambda/l$. The probability of absorption in the case of the potential distribution of type 1 is

$$\xi_0 = c \exp\left(-\frac{E_{a3}}{kT}\right), \tag{13}$$

where C is a constant (ca. 0.1–1). Substitution of b and c in equation (7) gives

$$\chi \approx \frac{c}{b}\frac{\lambda}{l}\exp\left(-\frac{E_{a3}}{kT}\right). \tag{14}$$

It is easy to transform this expression to the conventional one. Really, Figure 2 gives $E_{a3} = E_D + E_{ab}$, where E_D is the diffusion activation energy, and $E_{ab} \equiv E_{a3}$ is the absorption heat and, thus, $\chi \sim 1/l \exp(-E_D + E_{ab}/kT)$[6, 10]‡.

It follows from equation (14) that the gas permeability in the system with the potential distribution of type 1 can never exceed that of a hypothetic porous membrane $\chi = \lambda/l$. Obviously it is not reasonable to use a membrane for vacuum pumping in such a system.

In the case of the potential relief of type 2 (Figure 3), where $E_{a3} < 0$, things are different. There is no obstacle to gas absorption while absorbed particles in translational movement

* In the present paper we consider the system is at equilibrium in the case when the gas and membrane are at the same temperature and all the processes are of a thermal character (not stimulated from the outside), though different gas concentrations on both sides of a membrane make in this case the use of the notion 'equilibrium system' not quite strict (in particular, distribution of the particle concentration between the gas and the surface, the surface and the solid is not at equilibrium).

† According to reference 9 b and c (see equation (13)) in general depend on T according to a power law, however, in the present paper we shall neglect the power dependence on temperature.

‡ Note, that this conventional expression includes E_D and E_{ab} characteristics of a gas–solid system. But following our method of its deduction one can easily see that E_D and E_{ab} are included so far as they are constituents which comprise the boundary energy barrier on which ξ_0 depends (see equation (13)). It is sufficient to stimulate absorption (for example, by implanting ions) to make ξ_0 = constant and χ = constant, i.e. to eliminate E_D and E_{ab} dependence on permeability. We attract attention to this fact on purpose because usually not boundary, but bulk processes are connected with E_D and E_{ab} and direct interpretation of the equation (14) type formulas can result in incorrect conclusions on inevitable dependence of χ on E_D and E_{ab} (like, for example, the time of gas particle passage through the membrane depends on E_D).

can freely reach only the surface (position 1). From position 1, however, the motion back into solid is more probable than that into gas (reflection from a high surface—gas energy barrier takes place). Correspondingly, instead of equation (13) and $\zeta \approx 1$ here we get quite the reverse:

$$\xi_0 = c \approx 1, \tag{15}$$

$$\zeta_0 = \zeta_l = b \exp\left(-\frac{|E_{a3}|}{kT}\right). \tag{16}$$

In the high-temperature range

$$\left(T > \frac{|E_{a3}|}{k}\frac{1}{\ln l/\lambda b}\right)\zeta_0 = \zeta_l \gg \frac{\lambda}{l}$$

and according to equation (7)

$$\chi = \frac{c}{b}\frac{\lambda}{l}\exp\frac{|E_{a3}|}{kT} \tag{17}$$

when the temperature drops to

$$T < \frac{|E_{a3}|}{k}\frac{1}{\ln b(l/\lambda)}$$

strong internal reflection occurs $\zeta_0 = \zeta_l \ll \lambda/l$, which together with equation (15) ensures the trap and superpermeability

$$\chi \approx \tfrac{1}{2}\xi_0 \approx \tfrac{1}{2}. \tag{18}$$

Thus, in the case of the potential distribution of type 2 two interesting effects are possible: anomalous dependence of permeability on temperature, equation (17), in the high T-range and super permeability, equation (18), in the low T-range.

Now widespread and widely known are gas–solid systems with a potential distribution of type 1 (Figure 2) and little is known about those with a distribution of type 2 (Figure 3). Nevertheless, such systems exist. For example, apparently all equilibrium hydrogen–metal systems pertain to this type in the pressure and temperature range where a gas is dissociated into atoms. For example, a potential distribution of type 1 corresponds to a H_2–W system at $T < 1000$ K and at a pressure higher than 10^{-5} torr. However, the H_2–W system transforms to the H–W system at a pressure of 10^{-5} torr or at lower ones and $T > 2000$ K. Then the potential energy of an atom in a gas is raised by 2.2 eV* and the potential distribution of type 1 transforms to a potential distribution of of type 2. So it should be expected that an increase of the temperature in the range where hydrogen is already dissociated results not in the increase of the tungsten membrane gas permeability but in its decrease. Such a situation seems unusual and, nevertheless, in the first paper on hydrogen permeability of refractory metals in the high-temperature, low-pressure range on anomalous temperature dependence of the permeability has been obtained[11].

Unfortunately, it seems impossible to achieve in these particular systems supermeability by utilizing a potential distribution of type 2, because a required decrease of temperature transforms a H–Me system into a H_2–Me one and consequently the distribution of type 2 into that of type 1 (see Figure 3) with a usual gas permeability law, as has been observed[11].

* Note that in this case $kT \approx 0.2$ eV. A high level of the particle potential energy at a comparatively low temperature results from the change of the particle number during dissociation[12].

Thus, we can conclude that in known gas–solid equilibrium systems where a gas does not dissociate while being absorbed† only a very low gas permeability can occur which is not of interest in terms of gas evacuation.

5. Superpermeability in non-equilibrium systems in the case of absorption without dissociation

Now we consider the case when absorption is stimulated in a way that is

(1) independent of the membrane temperature and boundary conditions

$$\xi^* = c^* \approx 1 \tag{19}$$

(we shall designate ξ^* the probability of stimulated absorption);

(2) stimulation of absorption does not affect directly desorption, i.e. ζ_0 determines on the whole the membrane temperature.

Physically this means that now we are considering a gas–solid non-equilibrium system in which state distribution in a gas is shifted towards much higher energies from an equilibrium one corresponding to the membrane temperature. In this case condition (19) could be satisfied both by directly gas particle high energy and by particle special state (for example, excitation or ionization) for which boundary activation barriers may become relaxed or even eliminated.

As far as conservation of a desorption spontaneous character is concerned this can be always achieved, in particular by reducing the concentration when direct interaction of an absorbed particle with that being absorbed has a sufficiently low probability.

5.1. Low permeability on 'active' membranes. It is the result of absorption stimulation in the case of the potential relief of type 1 corresponding to our idea of an 'active' membrane (the surface is fully acceptable for absorption–desorption, the highest E_{a3} barrier is due to bulk material properties)? We can use, as before, expression (7), having substituted ξ_0 for ξ^* in it, as no limiting assumptions on the boundary process character have been made while deducing it. Substitution in equation (7) $\xi^* = c^*$ and $\zeta_0 = \zeta_l = b$ gives $\chi = c^*/b\ \lambda/l \approx \lambda/l$. Thus, absorption stimulation in type 1 systems results in permeation which is independent of the membrane temperature and material and close to that ultimately achieved by the increase of the membrane and gas temperatures (see equation (14)). This permeability does not exceed that of the hypothetical porous membrane with $d = \lambda$ and as has been already said it is too low to be used for evacuation. The reason for a low gas permeability is a backward motion of absorbed particles through the inlet boundary.

5.2. Superpermeability of 'inactive' membranes—the barrier trap. Let us raise the gas–surface barrier value E_{a1} by special surface treatment until E_{a1} becomes $> E_{a3}$. Then we get a potential distribution of type 3 instead of that of type 2 (see Figure 4). The treatment mentioned here corresponds to a membrane transition into an 'inactive' state.

In a system with a potential distribution of type 3 absorbed

† The equilibrium H–Me system in contrast to H_1–Me one pertains exactly to that type.

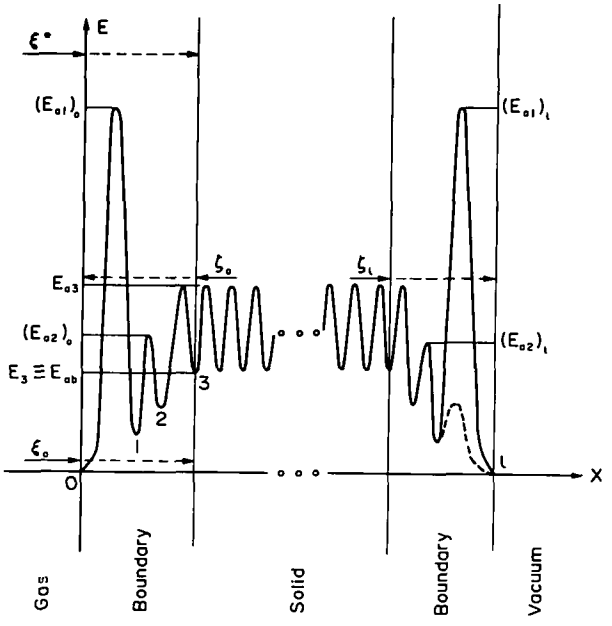


Figure 4. Potential distribution of a gas-membrane system of type 3: $E_{a1} > E_{a3}$. The type 3 distribution corresponds to the notion of a 'nonactivated' membrane.

particles 'cooled' to the membrane temperature upon reaching the high boundary barrier are reflected from it (as is the case with systems of type 2). The desorption probability at a single reaching of the boundary is here

$$\zeta_0 = \left[b \exp\left(- \frac{E_{a1} - E_{a3}}{kT} \right) \right]_0,$$
$$\zeta_l = \left[b \exp\left(- \frac{E_{a1} - E_{a3}}{kT} \right) \right]_l \qquad (20)$$

and consequently, by decreasing the temperature or increasing the boundary E_{a1} barrier, one can assure that the strong reflection conditions $\zeta_0 \ll \lambda/l$ are satisfied. However, in the case of interaction with a gas at equilibrium

$$\xi_0 = \left[c \exp\left(- \frac{E_{a1}}{kT} \right) \right]_0 \qquad (21)$$

and, thus, ξ_0 is reduced simultaneously with ζ_0. Substituting equations (20) and (21) in equation (7) we get (cf. equation (14))

$$\zeta_0 \gg \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l}, \quad \chi \approx \left(\frac{c}{b} \right)_0 \frac{\lambda}{l} \exp\left(- \frac{E_{a3}}{kT} \right),$$
$$\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l = \zeta_0, \quad \chi \approx \tfrac{1}{2} \left[c \exp\left(\frac{E_{a1}}{kT} \right) \right]_0$$
$$\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l \gg \zeta_0, \quad \chi \approx \left[c \exp\left(- \frac{E_{a1}}{kT} \right) \right]_0$$
$$\ll \left(\frac{c}{b} \right)_0 \frac{\lambda}{l} \exp\left(- \frac{E_{a3}}{kT} \right). \qquad (22)$$

In the stimulated absorption case the E_{a1} barrier does not prevent absorption, impeding desorption as before. Having assured, as has been indicated above, the strong reflection condition $\zeta_0 \ll \lambda/l$ we obtain a trap and superpermeability. In this case a considerable part of the incident gas flux is absorbed (C^* as an order of magnitude is comparable with 1) and a half (at $\zeta_l = \zeta_0$) or even all the absorbed particles (at $\zeta_l \gg \zeta_0$) permeate. Direct substitution of equations (19) and (20) in equation (7) gives

$$\zeta_0 \gg \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l}, \quad \chi \approx \left(\frac{c^*}{b} \frac{\lambda}{l} \exp \frac{E_{a1} - E_{a3}}{kT} \right)_0,$$
$$\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l = \zeta_0, \quad \chi \approx \tfrac{1}{2} c^* \approx \tfrac{1}{2},$$
$$\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l}, \quad \chi \approx c^* \approx 1. \qquad (23)$$

Conditions for ζ_0 and ζ_l in the left-hand side of equations (22) and (23) can be written explicitly using equation (20). Then it is easy to find corresponding ranges of E_{a1} and T variations at a given l value.

As is shown by the comparision of equations (22) and (23) the behaviour of the type 3 distribution system differs essentially in the cases of spontaneous and stimulated absorptions (see also Figure 5). In the ranges of temperatures and barrier

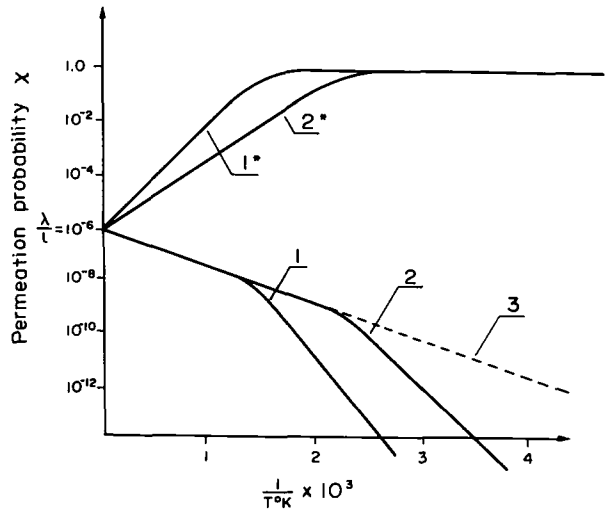


Figure 5. Example of temperature dependence of permeation probability in a system with the type 3 potential distribution at $E_{a3} = 0.25$ eV, $\lambda/l = 10^{-6}/l \approx 0.05$ cm. 1,2,—equilibrium conditions at $E_{a1} = 4E_{a3}$ and $E_{a1} = 3E_{a3}$ correspondingly; 1*, 2*—the same for a non-equilibrium gas; 3—equilibrium conditions at $E_{a1} < E_{a3}$ and $E_{a2} < E_{a3}$ (an activated membrane the type I relief). The curve bends correspond to transitions from a weak reflection to a strong one.

E_{a1} values, where $\zeta_0 \gg \lambda/l$, $\zeta_l \gg \lambda/l$ (weak reflection) conventional permeability decreases with decreasing temperature and it does not depend on the barrier E_{a1} value*. On the contrary, in the stimulated-absorption case the permeability increases with decreasing temperature and, moreover, it increases with the increase in the E_{a1} boundary barrier value. Upon achieving sufficiently low temperatures (or sufficiently high E_{a1} barriers) we come to the strong reflection conditions ($\zeta_0 \ll \lambda/l$), the usual permeability becoming much lower than that in the case of active membrane and falling off still faster with decreasing temperature (because $(E_{a1})_0 > E_{a3}$) and increasing the $(E_{a1})_0$ value of the boundary barrier. On the contrary, in the stimulated

* That is it remains as in the case of an 'active' membrane (cf. equation (14)).

absorption case the superpermeability is achieved, when χ^* becomes of the order of unity and depends neither on temperature nor on the boundary state (when the relation between ζ_0 and ζ_l remains unchanged).

Thus, all the factors resulting in the decrease of the conventional permeability result in the increase of the permeability for a non-equilibrium gas. The notion of the surface activation is also of the opposite sense in cases of a conventional and non-equilibrium permeability: namely, the 'inactive' low-permeable membrane with high boundary barriers becomes superpermeable in the case of stimulated absorption.

Traps based on the reflection of absorbed particles from the boundary barrier are called[1] 'barrier-type' ones. A barrier trap is a way to practical feasibility of superpermeability, that is, of the situation when for particles with energies of a few electron-volt capable of overcoming only one boundary barrier the permeability of a thick solid membrane appears to be comparable with that of an opening in a thin wall (including the case when the membrane temperature is low).

5.3. Superpermeability of 'inactive' membranes—a σ-trap. We can treat the pure 'active' surface of the membrane with a type 1 distribution in such a way that only the $\sigma \ll 1$ portion of the surface would be accessible for non-stimulated absorption–desorption (the remaining $(1 - \sigma)$ portion of the surface may, for example, be covered with 'a poison'—a foreign substance strongly bound to the surface). Assume that for absorption–desorption centres which remain free, a potential distribution of type 1 is true. Now desorption is impeded because of the lack of desorption centres and instead of $\zeta = b \approx 0.1$ 1, see equation (12'), we get

$$\zeta_0 = (\sigma)_0, \quad \zeta_l = (\sigma)_l. \tag{24}$$

The $\zeta_0 \ll \lambda/l$ condition for strong reflection can be satisfied by decreasing σ. In the case of conventional absorption ξ_0 decreases at the same time

$$\xi_0 = c_0\sigma_0 \exp - \frac{E_{a3}}{kT}, \tag{25}$$

which can result only in reduction of gas permeability (cf. equation (14))

$$\left.\begin{aligned} &\sigma_0 \gg \frac{\lambda}{l}, \quad \sigma_l \gg \frac{\lambda}{l}, \quad \chi \approx \frac{c_0}{b_0}\frac{\lambda}{l}\exp - \frac{E_{a3}}{kT}, \\ &\sigma_0 \ll \frac{\lambda}{l}, \quad \sigma_l = \sigma_0, \quad \chi \approx \tfrac{1}{2}c_0\sigma_0 \exp - \frac{E_{a3}}{kT} \\ &\sigma_0 \ll \frac{\lambda}{l}, \quad \sigma_l \gg \sigma_0 \quad\quad \chi \approx c_0\sigma_0 \exp - \frac{E_{a3}}{kT} \end{aligned}\right\} \ll \frac{c_0}{l_0}\frac{\lambda}{l}\exp - \frac{E_{a3}}{kT}. \tag{26}$$

Stimulated absorption proceeds through the $\sigma^* \approx 1$ part of the surface, which is practically constant while poisoning the surface*.

$$\xi^* = c^*\sigma^* \approx 1. \tag{27}$$

* Occupied parts of the surface can be considered, e.g. as those having a high activation barrier which cannot be really overcome in the case of conventional absorption–desorption, but its effect vanishes in the case of absorption of the non-equilibrium gas.

With this we get from equations (24), (27) and (7)

$$\begin{aligned} &\sigma_0 \gg \frac{\lambda}{l}, \quad \sigma_l \gg \frac{\lambda}{l}, \quad \chi^* \approx \frac{c^*}{b_0}\frac{\sigma^*}{\sigma_0}\frac{\lambda}{l} \approx \frac{1}{\sigma_0}\frac{\lambda}{l}, \\ &\sigma_0 \ll \frac{\lambda}{l}, \quad \sigma_l = \sigma_0, \quad \chi^* \approx \tfrac{1}{2}c^*\sigma^* \approx \tfrac{1}{2}, \\ &\sigma_0 \ll \frac{\lambda}{l}, \quad \sigma_l \gg \sigma_0, \quad \chi^* \approx c^*\sigma^* \approx 1. \end{aligned} \tag{28}$$

Thus, in contrast to an ordinary gas permeability, equation (26), stimulated-absorption gas permeability, equation (28), increases while the surface is being poisoned until the strong reflection conditions are satisfied and superpermeability is achieved which is independent of the surface condition (remember that always $\sigma^* \approx 1$). A so-described trap is called in reference 1a σ-trap. As can be seen from equation (28) construction of the trap this way is characterized by the absence of T dependence of the permeability in the whole region of χ^* variation (cf. equation (23)).

The increase of the boundary barrier up to $E_{a1} > E_{a3}$ and reduction of the number of desorption centres may be obviously achieved simultaneously (see Experimental Part). In this case the superpermeability is achieved by the combination of the barrier- and σ-traps. Corresponding formulas are easily obtained

$$\begin{aligned} &\xi^* = c^*\sigma^* \approx 1, \\ &\xi_0 = \left[c\sigma \exp\left(-\frac{E_{a1}}{kT}\right)\right]_0, \end{aligned} \tag{28'}$$

$$\begin{aligned} &\zeta_0 = \left[\sigma \exp\left(-\frac{E_{a1} - E_{a3}}{kT}\right)\right]_0, \\ &\zeta_l = \left[\sigma \exp\left(-\frac{E_{a1} - E_{a3}}{kT}\right)\right]_l, \end{aligned} \tag{29}$$

$$\left.\begin{aligned} &\zeta_0 \gg \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l}, \quad \chi \approx \frac{0}{b_0}\frac{\lambda}{l}\exp\left(-\frac{E_{a3}}{kT}\right), \\ &\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l = \zeta_0, \quad \chi \approx \tfrac{1}{2}\left[c\sigma \exp\left(-\frac{E_{a1}}{kT}\right)\right]_0 \\ &\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l \gg \zeta_0, \quad \chi \approx \left[c\sigma \exp\left(-\frac{E_{a1}}{kT}\right)\right]_0 \end{aligned}\right\} \ll \frac{c_0}{b_0}\frac{\lambda}{l}\exp\left(-\frac{E_{a3}}{kT}\right), \tag{30}$$

$$\begin{aligned} &\zeta_0 \gg \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l}, \quad \chi^* \approx \left(\frac{c^*\sigma^*}{\sigma}\exp\frac{E_{a1} - E_{a3}}{kT}\right)_0\frac{\lambda}{l}, \\ &\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l = \zeta_0, \quad \chi^* \approx \tfrac{1}{2}c^*\sigma^* \approx \tfrac{1}{2}, \\ &\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l \gg \zeta_0, \quad \chi^* \approx c^*\sigma^* \approx 1. \end{aligned} \tag{31}$$

The range of σ, E_{a1}, T and l values where one or the other expression for χ or χ^* is true are obtained by direct substitution of equation (29) into conditions written in the left hand side of equations (30) and (31).

6. Solid membrane superpermeability in the case of absorption with dissociation

Dissociation into atoms or ions occurs, as is well-known, during adsorption and absorption of hydrogen by metals and, probably, in other diatomic gas–metal systems.

6.1. Superpermeability in a system at equilibrium†. As has been shown above, in the absence of dissociation the probability of permeation of particles of an equilibrium gas through the potential distributions of types 1 and 3 can only tend to the λ/l value. A low ultimate permeability results from the fact that in the case of interaction with a gas at equilibrium one cannot satisfy simultaneously two conditions $\xi_0 \approx 1$ and $\zeta_0 \ll \lambda/l$ because reduction in the desorption probability causes inevitably reduction in the absorption probability.

The situation is different in the case of absorption accompanied by dissociation. Here, in the wide temperature range spontaneous desorption of gas particles can occur only by pairs—by means of their recombination on the surface.

Reduction of the concentration impeding desorption from the surface results in two effects: on one hand the probability ζ_0 of the absorbed-particle transition into the gas (3–0) transition (Figures 2–4) is reduced because having reached the surface and not having found a match the particles return back into the solid; on the other hand, the absorption probability ξ_0 increases (0–3 transition) because particles that have just been adsorbed also transfer to the solid not having found a match to desorb at the surface. In particular, one can always indicate a rather low concentration in the gas at which practically each adsorbed particle is also absorbed including the case of the potential distribution of type 1. Further reduction of the concentration not reducing the absorption probability will cause continuous reduction of the probability ζ_0 of return to the gas and consequently continuous increase of the permeation probability. At a sufficiently low concentration the strong reflection will occur and the probability of permeation will become of the order of that of adsorption. Thus, in the case of absorption with dissociation the limit on the χ value is not the λ/l value but the adsorption probability (sticking coefficient). The adsorption probability can achieve the value of ≈ 1 for diatomic gases on pure metal surfaces including a low-temperature case. For example, it is known that hydrogen chemisorption on the Ni, Ti, Zr, Nb, Ta, Cr, Mo, W, Fe, Co, Rh, Pd, Pt pure surface occurs without an activation energy[13]. Thus, a solid membrane can be superpermeable for an equilibrium gas if the latter dissociates in the process of absorption.

This possibility has been qualitatively considered in reference 1. According to this work at rather a low concentration in a gas (see Figure 1 and any of Figures 2–4)

$$\xi_0 = \left[c\sigma \exp\left(-\frac{2E_{a1}}{kT}\right)\right]_0, \tag{32}$$

$$\zeta_0 = \frac{1}{G\alpha}\left[c\sigma \exp\left(-\frac{2E_{a1} - E_{a3} - E_3}{kT}\right)\right]_0 n_{ab}(0), \tag{33}$$

$$\zeta_l = \frac{1}{G\alpha}\left[c\sigma \exp\left(-\frac{2E_{a1} - E_{a3} - E_3}{kT}\right)\right]_l n_{ab}(l). \tag{34}$$

† See note on page 107.

In equations (32)–(34) σ is, as before, a part of the surface accessible for non-stimulated absorption–desorption, α—the constant determined from an equation (see Figure 2), (remember that v_a and v are mean translational velocities of particles in the membrane and in the gas), G—the constant determined from the equation (see Figure 2)

$$\frac{n_{abe}}{n} = G \exp\left(-\frac{E_3}{kT}\right), \tag{36}$$

where n_{abe} is the equilibrium concentration in a membrane material.

Expression (32) means that each adsorbed particle is absorbed, and equations (33) and (34) mean that the desorption process is of the second order in regard to absorbed-particle concentration near the boundary. Now it is a more difficult task to reduce equation (7) to χ explicit form, because ζ_0 and ζ_l depend on $n_{ab}(0)$ and $n_{ab}(l)$, and these are determined in their turn from equation (6). However, in the limiting cases of strong and weak reflection of interest in the present work, a simple expression can be obtained for the probability of permeation χ:

$$\begin{aligned}
&\zeta_0 \gg \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l}, \quad \chi \approx \sqrt{G\alpha}\,\frac{\lambda}{l}\frac{1}{\sqrt{n}} \exp\left(-\frac{E_{a3}}{kT}\right), \\
&\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l = \zeta_0, \quad \chi \approx \tfrac{1}{2}\xi_0 = \tfrac{1}{2}\left[c\sigma \exp\left(-\frac{2E_{a1}}{kT}\right)\right]_0, \\
&\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l \gg \zeta_0, \quad \chi \approx \xi_0 = \left[c\sigma \exp\left(-\frac{2E_{a1}}{kT}\right)\right]_0.
\end{aligned} \tag{35}$$

The condition of transition from the first line of equation (35) to the second or third one ($\zeta_0 \ll \lambda/l$) can be written in the explicit form[2]

$$\sqrt{G\alpha}\,\frac{\lambda}{l}\frac{1}{\sqrt{n}} \exp\left(-\frac{E_{a3}}{kT}\right) \gg \left[c\sigma \exp\left(-\frac{2E_{a1}}{kT}\right)\right]_0. \tag{36}$$

Choosing sufficiently low values of n this condition can be satisfied at *any* values of E_{a1}, T and σ including the case

$$\left[\sigma \exp\left(\frac{2E_{a1}}{kT}\right)\right]_0 \approx 1$$

when, according to equation (35), $\chi \approx \frac{1}{2}c_0 = \frac{1}{2}(0.1 - 1)$ or $\chi \approx c_0 = 0.1 - 1$‡. So, in the case $\zeta_0 - n_{ab}(0)$ (formula (33)) we can obtain a concentration trap and superpermeability if

$$T > \left(\frac{2E_{a1}}{kT}\right)_0, \quad \sigma_0 = 1, \tag{37}$$

$$n \ll \frac{G\alpha^2}{C_0^2}\frac{\lambda^2}{l^2} \exp\frac{2E_{a3}}{kT}. \tag{38}$$

If conditions (37) and (38) are satisfied but the condition $\zeta_0 \sim n_{ab}(0)$ is not yet satisfied then to assure superpermeability

‡ For comparison in the case (30) the condition $\zeta_0 \ll \lambda/l$ is analogous to equation (36), where ζ_0 is taken from equation (29).

$$\frac{1\lambda}{l} \exp\left(-\frac{E_{a3}}{kT}\right) \gg \left[\sigma \exp\left(-\frac{E_{a1}}{kT}\right)\right]_0. \tag{36*}$$

In contrast to equation (36) the right hand side of equation (36*) is always less than λ/l, consequently $\sigma \exp - E_{a1}/kT \ll \lambda/l \ll 1$ is always true, consequently (see equation (30)) $\chi \ll \lambda/l \ll 1$ is also true.

additional reduction of n is necessary until $\zeta_0 \sim n_{ab}(0)$. In reference 2 the conditions are stated under which relations (32)–(34) exist.

For example, for a H_2–Ni system $E_{a1} = 0^{13}$, $E_{a3} = 14$ kcal/g atom[14], $G \approx 1.5 \times 10^{21}$ atoms/cm^3 [17], $\alpha = 1.6$. Assuming $c_0 = 0.1 - 1$ one can expect that for a 1×10^{-2} cm thick pure surface membrane the superpermeability would occur in the range $n \ll 6 - 600$ molecules/cm^3 (2×10^{-16}–10^{-14} torr) at $T = 600$ K and in the range $n \ll 10^{-19}$ molecules/cm^3 ($\approx 10^{-35}$ torr) at $T = 300$ K (or at still lower n if at n values attained above the dependence $\zeta_0 \sim n_{ab}(0)$ is not yet reached).

For a H_2–Pd system $E_{a3} \approx 4$ kcal/g atom, $G \approx 1.5 \times 10^{21}$ atoms/cm^3, $\alpha \approx 1.6$[14]. Assuming c_0 0.1–1 and $E_{a1} = 0^{13}$ for a 1×10^{-2} cm thick pure surface membrane one can expect that superpermeability would occur in the concentration range below 4×10^6–10^8 molecules/cm^3 (10^{-10}–10^{-8} torr) at $T =$ 800 K (see Experimental Part) and 10^3–10^5 molecules/cm^3 (3×10^{-13}–10^{-11} torr) at 300 K.

Finally, a H_2–$Pd_{0.75}$–$Ag_{0.25}$ system differs from a H_2–Pd one mainly in that $E_{a3} \approx 2$ kcal/g atom. Consequently the concentration range sought for H_2–$Pd_{0.75}$–$Ag_{0.25}$ may exist below 5×10^7–10^9 molecules/cm^3 (10^{-9}–10^{-7} torr) at 800 K and 6×10^5–10^7 molecules/cm^3 (2×10^{11}–10^9 torr) at 300 K.

6.2. Superpermeability in the case of a non-equilibrium gas. In the case of stimulated absorption instead of equation (32) one will have $\xi^* = c^*\sigma^* \approx 1$ while equations (33) and (34) remain the same. In this case

$$\zeta_0 \gg \frac{\lambda}{l}, \quad \zeta_l \gg \frac{\lambda}{l},$$

$$\chi^* \approx \sqrt{G\alpha} \sqrt{\frac{c^* v}{c_0 v^*}} \frac{\lambda}{l} \sqrt{\frac{\sigma^*}{\sigma_0}} \frac{1}{\sqrt{n}} \exp \frac{E_{a1} - E_{a3}}{kT},$$

$$\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l = \zeta_0, \quad \chi^* \approx \tfrac{1}{2}\xi^* = \tfrac{1}{2}c^*\sigma^* \approx \tfrac{1}{2},$$

$$\zeta_0 \ll \frac{\lambda}{l}, \quad \zeta_l \gg \zeta_0, \quad \chi^* \approx \xi^* = c^*\sigma^* \approx 1. \tag{39}$$

Now, if $\zeta_0 \sim n_{ab}(0)$ is true, to achieve superpermeability it is sufficient only to provide conditions of strong reflection $\zeta_0 \ll \lambda/l^2$.

$$n \ll G\alpha^2 \frac{v}{v^*} \frac{\lambda^2}{l^2} \left[\frac{1}{c_0 c^* \sigma_0 \sigma^*} \exp \frac{2(E_{a1} - E_{a3})}{kT} \right]_0. \tag{40}$$

Thus, superpermeability may be achieved for all surface conditions (in particular, when $\sigma_0 \ll 1$ and $E_{a1} > E_{a3}$) and for any temperature. Because σ, E_{a1} and T values are now arbitrary in contrast to equations (37) and (38), one can, according to equation (40), obtain superpermeability by increasing E_{a1}, reducing σ and T (at $E_{a1} > E_{a3}$) in the range of concentrations and thicknesses higher than those in the case of equilibrium gas. The trap which is formed is not a 'pure' concentration one but a mixture of all the three types of traps: a barrier trap (at $E_{a1} > E_{a3}$), a σ-trap and a concentration trap. As follows from the first line of equation (39), together with satisfaction of the conventional law $\chi \sim 1/l\sqrt{n}$ in the case of stimulated absorption the anomalous temperature dependence of permeability (at $E_{a1} > E_{a3}$) and enhancement of permeability in the process of surface passivation (reduction of σ and increase of E_{a1}) can be observed. Thus, as in the case of absorption without dissociation, the same changes in the system lead to different results in the cases of equilibrium and non-equilibrium gases.

7. Gas compression

Consider requirement (2) stated in the Introduction of the present paper. Let the gas be on the right and on the left sides of the membrane (Figure 6) in stationary (but not necessarily

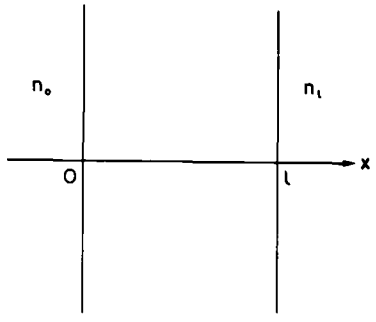


Figure 6. Consideration of gas on either side of a membrane.

equilibrium) condition. If the system is closed, stationary n_0 and n_l concentrations are established on both sides of the membrane. These concentrations can be determined from the condition of absence of flux through the membrane: $v_0 n_0 \chi_0 = v_l n_l \chi_l$, where χ_0 and v_0 are the probability of permeation during a single strike and the mean velocity on the left hand side, and χ_l and v_l are the same values on the right hand side. Thus, the possible compression ratio is

$$\beta = \frac{n_l}{n_0} = \frac{v_0 \chi_0}{v_l \chi_l}. \tag{41}$$

For such a compression ratio the rate of gas evacuation is equal to 0. If we would like to use completely the membrane permeability for gas evacuation, the compression ratio should be several times less than β.

7.1. Compression in the case of a non-equilibrium gas. It goes without saying that if stimulated absorption which increases the permeability is realized only on one side of the membrane, the gas passing through the membrane can be compressed in some way. The fact is less evident but extremely favourable that permeability of an equilibrium and non-equilibrium gas appears to be, as has been shown earlier, inversely dependent on the degree of surface activation (σ, E_{a1}) and the membrane temperature.

In the process of absorption without dissociation the superpermeability can be achieved in general only when permeability of an equilibrium gas is suppressed to a great extent. Remember, that the maximum value of the latter is extremely low: ($\chi \approx (\lambda/l) \exp -E_{a3}/kT$).

In the case of absorption with dissociation the superpermeability of a non-equilibrium gas is also achieved in the range of concentrations, temperatures and membrane thicknesses, which is the widest when the usual equilibrium permeability is suppressed to the greatest extent, however, in principle it is possible here to achieve simultaneously the superpermeability for equilibrium and non-equilibrium gases when it is

impossible to achieve high compression ratio due to stimulated absorption.

For a gas with a state distribution corresponding to the membrane temperature the membrane permeability in both directions is the same when membrane boundaries are both identical and non-identical.

For this reason χ_l in equation (41) is simply χ (see equation (7)). Further, $v_0 = v^*$, $\chi_0 = \chi^*$. As, in accordance with equation (7), $\chi^*/\chi = \xi^*/\xi_0$ we get

$$\beta = \frac{v^*}{v_l}\frac{\xi^*}{\xi_0}. \tag{42}$$

According to equation (42) β depends neither on the membrane thickness nor on the state of the membrane outlet boundary. In case of absorption without dissociation substituting in equation (42) ξ_0 from equation (28′) and $\xi^* = c^*\sigma^* \approx 1$ we get

$$\beta = \frac{v^*}{v_l}\frac{c^*\sigma^*}{c\sigma}\exp\frac{E_{a1}}{kT} \approx \frac{v^*}{v_l}\frac{1}{\sigma}\exp\frac{E_{a1}}{kT}. \tag{43}$$

If superpermeability conditions are achieved for a non-equilibrium gas the strong internal reflection condition is satisfied. $\zeta_0 \ll \lambda/l$. According to equation (29) this means that

$$\frac{1}{\sigma}\exp\frac{E_{a1}}{kT} \gg \frac{l}{\lambda}\exp\frac{E_{a3}}{kT}$$

and consequently $\beta \gg l/\lambda \exp E_{a3}/kT$. According to equation (43) this high compression ratio can be still increased if necessary without any loss in permeability by the surface passivation (by increasing E_{a1} and reducing σ) and reducing T (at $E_{a1} > E_{a3}$).

In case of absorption with dissociation the compression ratio with due account of equation (32) is

$$\beta = \frac{v^*}{v_l}\frac{c^*\sigma^*}{c\sigma}\exp\frac{2E_{a1}}{kT} \approx \frac{v^*}{v_l}\frac{1}{\sigma}\exp\frac{2E_{a1}}{kT}$$

and it can be made of a required value also without any loss in superpermeability for a non-equilibrium gas by surface passivation and reducing the temperature (see Experimental Part).

7.2. Compression in the case of an equilibrium gas. As has been said in Section 6.1. high permeability can be achieved for an equilibrium gas provided the gas dissociates in a solid. However, the necessary gas compression for evacuation is less evident here than in the case of stimulated absorption and achievment of compression seems to be specific for any particular case. For example, Young[15] has found that if a palladium membrane separates the volume filled with hydrogen from the atmosphere then the hydrogen pressure would be reduced due to hydrogen permeation through the membrane the final value being not 4×10^{-4} torr (H_2 pressure in the atmosphere) but 10^{-9} torr. Young indicates that the reason for such 'an anomalous diffusion' is catalytic transformation of hydrogen into water at the palladium surface in an oxidizing medium, due to the high oxygen pressure the hydrogen concentration at the outlet membrane surface corresponds to its pressure of 10^{-9} torr. Being transformed into water and desorbed the gas passing through the barrier is compressed from 10^{-9} torr to the water partial pressure in the atmosphere, i.e. approximately by a factor of 10^{11}.

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